

Hemigraspus oregonensis, Hairy Shore crabs, are a species of small, intertidal crabs that live in the Pacific Northwest. Unfortunately, they are under threat by an invasive predator. *Carcinus maenas*, European Green Crabs, are native to waters near the UK but were accidentally brought to the PNW. With no natural predators, they are able to diminish populations of the much smaller shore crabs without being kept in check. Additionally, human activity has polluted the waters of the PNW, causing natural systems to become unbalanced. This could potentially mean that the pollutants in the water could contribute, in some regards, to the decline of hairy shore crab populations as a result of European green crab invasion. Here we show that pollution does affect the reactions of shore crabs to green crabs. In using, Resazurin® and measuring antennae flicking, we determined the effect pollution has on the shore crabs. The results of the Resazurin showed that metabolic rates when exposed to just predator pheromones increased as expected, but were nominal when the same pheromones were injected into a jar of polluted water. For antenna flicking, a stress proxy, there was no increase in flick speed in polluted water when exposed to predator pheromones and pollution, suggesting that the crabs in polluted water couldn't detect the predator was there. Our results demonstrate that pollution is inhibiting the ability to smell the green crabs before the shore crabs are eaten. Without clearly smelling the green crabs, the shore crabs have no way of viably escaping since they can't rely on sound or sight in the water. We anticipate these findings to be a jumping off point, so to speak for climate scientists to further research into human impacts of the PNW. The results of this experiment can't definitively prove if the human impact on the environment has lead to a direct increase in green crab predation of shore crabs, that this is universal for the entire PNW, nor how European green crabs are affected by the pollution. However, this study does find that there is a potential relationship between crab behavior and pollution of the water that warrants further research to determine.

Introduction

Hairy shore crabs (*Hemigraspus oregonensis*) are one of the most abundant native crabs in Washington State and the greater Pacific Northwest. They're found in tidepools and mudflats along the coast, feeding on algae and providing food for shorebirds and other native predators. The key word being, native. "An introduced population of European green crabs *Carcinus maenas* was established in...southern Oregon (approximately 1996), the Oregon and Washington coastal estuaries and the west coast of Vancouver Island, British Columbia (1998)...and into the inland Salish Sea in 2015 and 2016." (Yamada et al 2021). Despite the high number of crabs currently in the tidepools of Washington and Oregon, the populations are declining due to green crab predation. "In some regions, including the Atlantic coast of North America, *C. maenas* has caused long-term degradation of eelgrass habitats and bivalve, crab, and finfish populations" (Ens et al 2022). The waters of the Pacific

Northwest are so ideal for the European Green crabs that they reproduce rapidly, faster than environmental scientists are able to ethically remove them from the area. However, this could change for both crab species as the waterways in coastal regions of the Salish Sea become more polluted by large urban sprawl like Seattle. To see what kind of effect pollution had on these crabs, we devised an experiment to test stress proxies of the shore crabs in response to the presence of European Green crabs. We measured the metabolic rate and the antenna flick rate of the crabs over 2 weeks of observation. We tested the null hypothesis that the pollution wouldn't have any effect on the crabs. We predicted that the pollution would affect the detection ability of the hairy shore crabs. More specifically, the shore crabs wouldn't be able to detect the green crabs as well in polluted waters as they do in non-polluted waters.

Methods

We tested 8 *Hemigraspus oregonensis* and 4 *Hemigraspus nudus*, dividing them up equally 4 containers. We also filled 2 tubs with saltwater collected from Puget Sound. In one of the tubs, we placed a concoction of copper paint, herbicide, and tire flakes from the street outside the lab in order to simulate polluted waters. Then, 2 open aired, labeled jars of 3 crabs, 1 nudus and 2 oregonensis were weighed for their mass and placed in each of the tubs. The crabs were left to acclimate to their environments for about 1 week. On week 2 of the experiment, 2 specified jars, one from the nonpolluted water and one from the polluted water were placed for 10 minutes in an equal tub but with a singular European Green crab, with pheromones able to be smelled by the shore crabs with some additional pipetting of water surrounding the European Green Crab and injected straight into the jar of shore crabs in order to ensure pheromones could be detected. After those 10 minutes, each of the jars were extracted from their tubs and the antenna flicking rate of each of the crabs were recorded with a clicker counter. While the crabs' antennae were being observed, a Resazurin® solution was created by first preparing the stock solution into a 6 wellled tray. Each crab from both jars in 1 tub were placed into a singular well and left to respire for 30 minutes. At the same time, select concentrations were loaded into the far side of a 96 well plate, labeled blanks. 1 blank was made for every 2 wells in order to ensure that when incubated, accurate results are achieved. After 30 minutes passed, 20 μmol of the solution the crabs had respired in was transferred to the 96 well plate. The crabs were then left to respire for another 30 minutes and then 20 μmol of that solution was also loaded into the 96 well plate. The crabs were then placed back into their standard tubs and left to acclimate another week. In week 3 of the experiment, the same process was repeated with the antennae flicking and Resazurin®. Once all data had been collected, the crabs exposed to pollution were quarantined in a specific baggie and placed in a freezer for human euthanasia. The

shore crabs exposed to non-polluted water were then placed back with the larger sample of collected shore crabs. Resazurin® was then incubated and metabolic rates were calculated and graphed, along with the antennae flicking rates.

Results

After incubation of the Resazurin® samples, a box and whisker plot was created to account for slight variation during the incubation process. The fluorescence of the control, the jar exposed to no predators and nonpolluted water showed a slight uptick in fluorescence that could likely be attributed to the small space the 3 crabs were in, but for argument sakes, it was zero. For predator cues, the fluorescence was around 2.5 RFU with outliers at 2.9 and 0.7 RFU. For the polluted samples, we saw that the jar not exposed to predator pheromones were much lower than 1 RFU. For the polluted sample that was exposed to predators, we saw roughly the same RFU as the polluted sample, but slightly higher in some points. For the antenna flicking, our control jar was roughly at 110 +/- 20 flicks per min for each crab. The jar only exposed to predator pheromones ended up being roughly 85 +/- 20 flicks per min for each crab, lower than the control. For the polluted samples, the jar not exposed to predators was at roughly the same as the control except in the case of high pollution wherein it dropped to roughly exponentially to 5 flicks per min. In the polluted sample exposed to predators, the flicking was also quite similar to the control samples value at 110 +/- 35 flicks per min.

Discussion

It is important to make this disclaimer. After we had left the crabs to acclimate to their new environment for a week, when we collected them in the 2nd week of the experiment, one of the jars exposed to the polluted water had killed all the crabs. We hadn't adjusted anything with the water aside from the mixture of pollutants we added. We hypothesized that the concoction of chemicals had been too much and too concentrated for the crabs to handle and they had died. So, we were forced to adjust the experiment slightly and obtain a new crab and do a hastened acclimatization to achieve nondisruptive data. The results still ended up being within acceptable ranges, but it should be noted that repeats of this experiment should be performed in order to validate said results. For the flicking, human error could have easily caused miscounts. Rapid movement not fully being digested by the brain as well as debris in the water could have skewed data. An experiment always demands replication, but in this case, replication is essential to ensure that data is not skewed. As for the overall results, it seems we are able to reject the null hypothesis. The pollution did have an effect on the hairy shore crabs. We saw that in the polluted samples, the metabolic rates in hairy shore crabs exposed to predator pheromones were much lower, similar to crabs that aren't detecting the European Green crabs nearby. As for antenna flicking, the flicks were almost the same

as the control in polluted samples regardless of predator presence. This means that it's not able to smell the European Green Crab. The only fluke, the 5 flicks per min is likely due to the hairy shore crab not wanting to smell the putrid water around it, which is also problematic. If the crab doesn't want to smell, it won't be able to effectively detect its predator. What this infers is something potentially destructive, like in the near collapse of the New England soft shelled clam fishery in the 1940's also due to the European Green Crab's. "The New England landings of soft-shell clams, *Mya arenaria* L., declined from a high of 14.7 million pounds in 1938 (Fiedler, 1940) to a low of 2.3 million pounds in 1959 (Power, 1961)" (Ropes 1968). While it's clear that clams and crabs aren't the same thing, the idea that invasive species with high adaptability and voracious appetites can almost wipe out several million clams in 20 years is almost unfathomable. The effects can be translated into predictions about the future of *Hemigrapsus oregonensis*, in that the likely event will be the destruction of the species. Now with the ability to detect their predators and not having the sense to escape, the logical conclusion is a potential increase in predation in the wild. For intertidal crabs in the PNW, the environment is becoming even more hostile. With local climates changing drastically the threat of storms disrupting the ecosystem looms greater over the coastal regions of the Salish Sea. Especially considering El Nino has a significant role in Green Crab presence. "The ranks and filters were integrated to determine the relative contribution of each source of *C. maenas* to risk...during an El Nino year. The results suggest that the habitat and endpoint with the greatest risk are the eelgrass habitat and the juvenile Dungeness crab, respectively." (Colnar and Landis 2005). Again while not directly hairy shore crabs, the results can be extrapolated and applied. Climate change is going to start changing the weather patterns, which means El Nino can becoming more frequent and erratic. Coupled with the already polluted water, the storms increasing *C. maenas* density in PNW spells out a severely negative fate for the hairy shore crabs. The most unfortunate part is that there isn't a viable solution to this problem. The best way to combat this issue is to further research on effects of European Green crabs onto Hairy Shore Crabs.

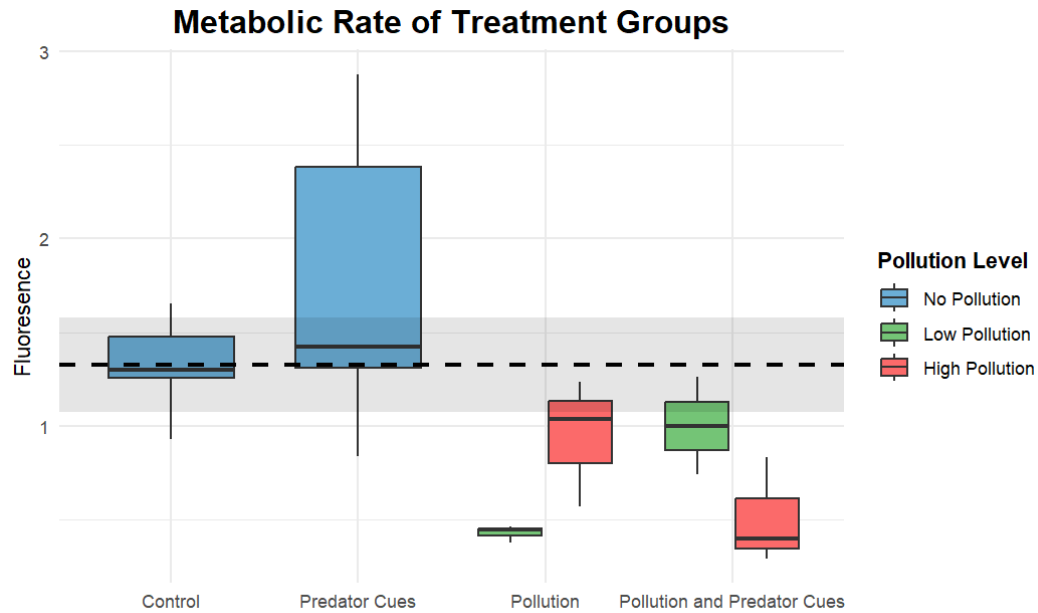


Figure 1. The average metabolic rates of 3 crabs per jar under 4 different conditions, control (no pollution no predators), exposure to just predators, exposure to just pollution, and exposure to predators and pollution. Crab were placed in Resazurin® solution for 30 minutes and 20µl samples were extracted from the solution and incubated to achieve results

Mean Antennae Flicking by Treatment and Pollution Level

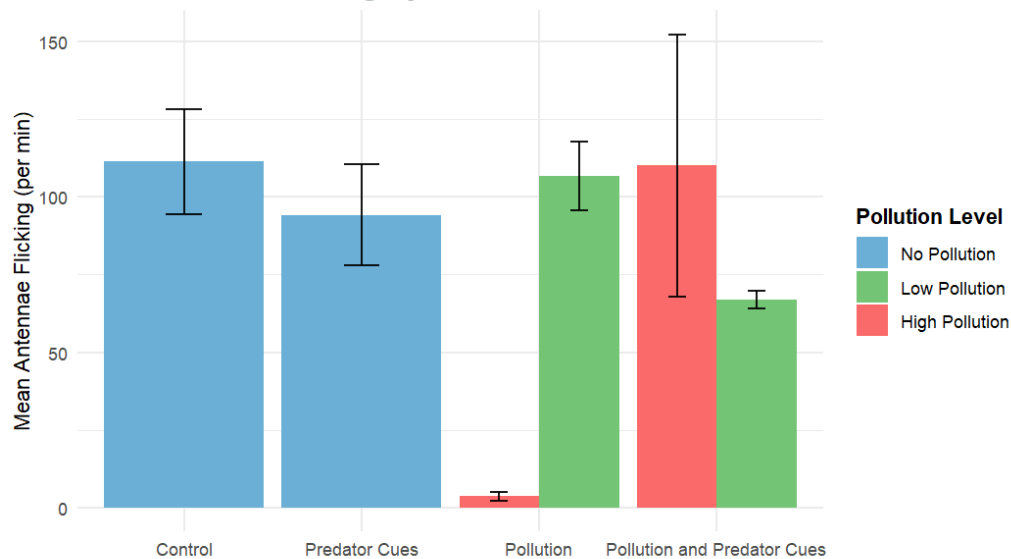


Figure 2. The average flicking rate of hairy shore crabs under 4 different conditions, control (no pollution no predators), exposure to just predators, exposure to just pollution, and exposure to predators and pollution. Each crab was observed and had their antennae flicking recorded for 1 minute.

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